



When Does Verification Pay Off?

A Closer Look at LLMs as Solution Verifiers

Jack Lu* Ryan Teehan* Jinran Jin Mengye Ren



Problem Setting

Research Question

When does an LLM verifier improve a solver at test time?

Evaluation Scope

37 models, 4 families, and 9 benchmarks across self-, intra-family, and cross-family verification settings.

Main Findings

Cross-family verification outperforms self-verification; solver-verifier similarity hurts improvement; verifier gain is more useful than verifier accuracy; task type matters; post-training reduces self-improvement.

New Metric: *Verifier Gain*

$$\underbrace{Gain(S, V; D)}_{\text{performance improvement from using verifier}} = \underbrace{Precision(S, V; D)}_{\text{accuracy after test-time rejection sampling with verifier}} - \underbrace{SolverAccuracy(S; D)}_{\text{base solver's accuracy}}$$

Verifier gain measures how much a solver improves when a verifier filters its sampled outputs through rejection sampling.

Experimental Design

Rejection Sampling

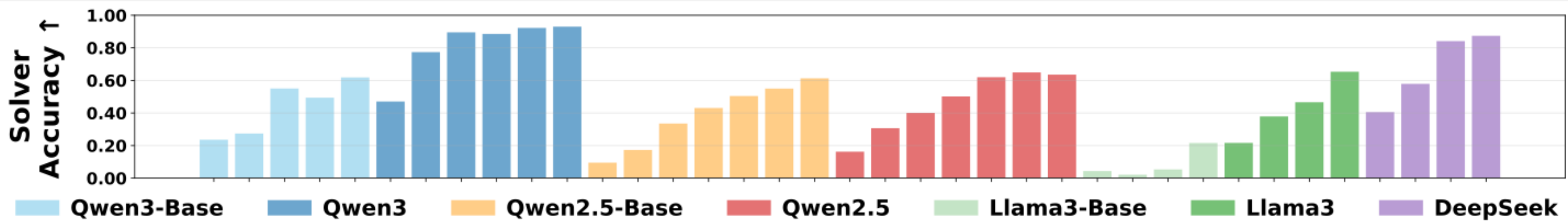
The solver generates one solution at a time. The verifier accepts or rejects it. Sampling stops once an answer is accepted or the budget is exhausted.

Improvement Settings

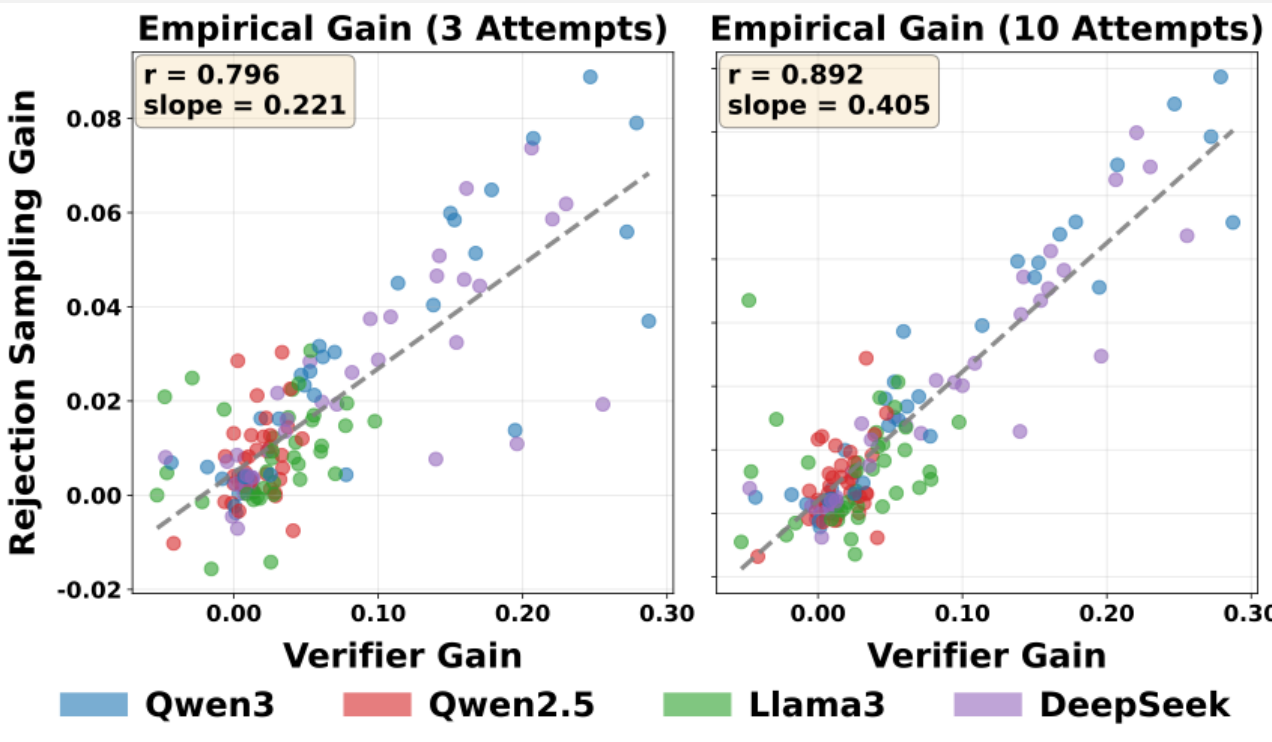
- **Self-improvement:** Can a model improve its own solutions?
- **Intra-family improvement:** Can one model improve solutions from another model in the same family?
- **Cross-family improvement:** Can one family improve solutions generated by another family?

#1 Solver Accuracy Rises With Model Scale

Solver performance improves consistently with scale across families, with Qwen3 and DeepSeek emerging as the strongest overall. Models within each family are ordered by increasing size.



#2 Verifier Gain Predicts Real Test-Time Improvement



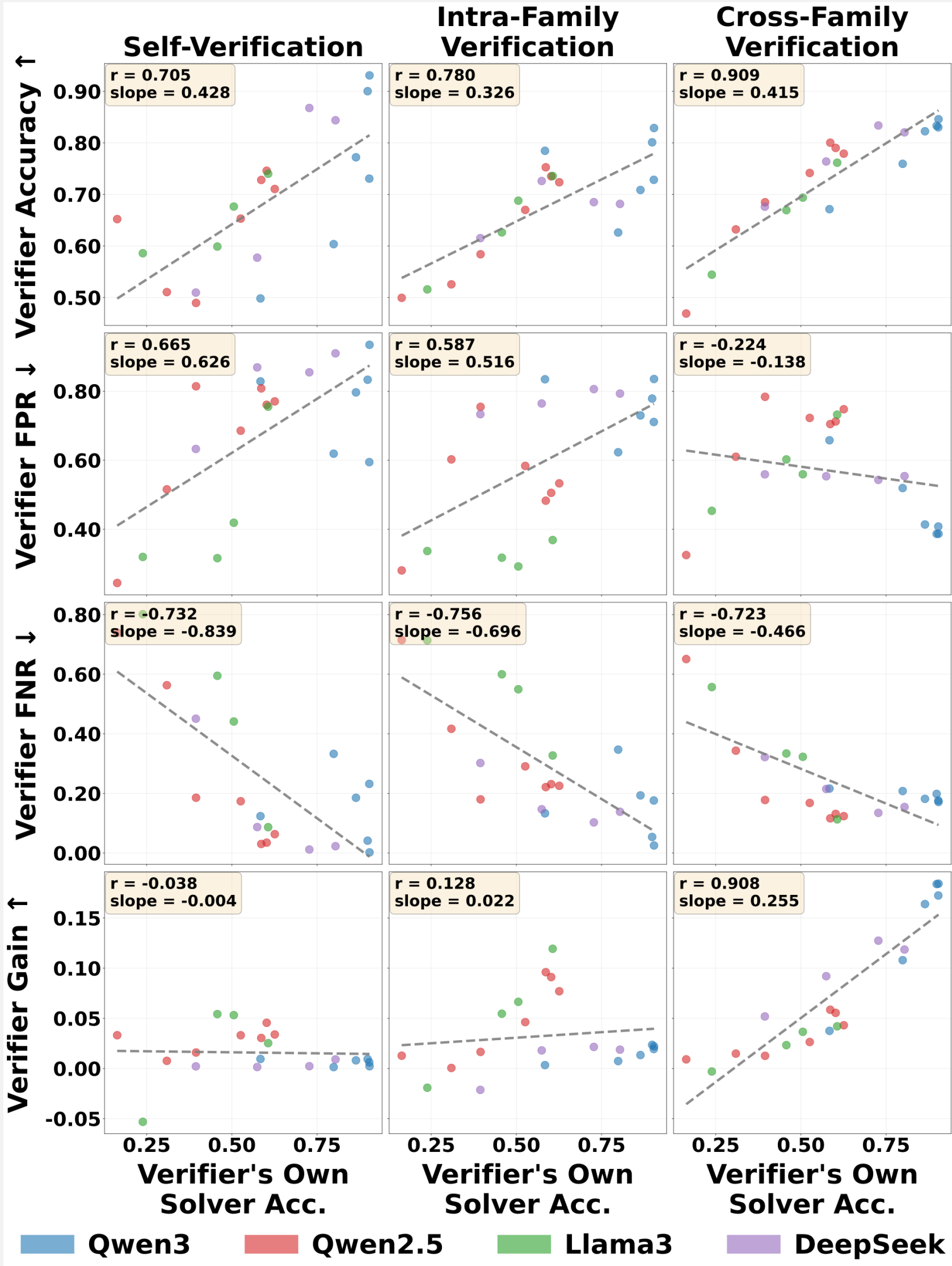
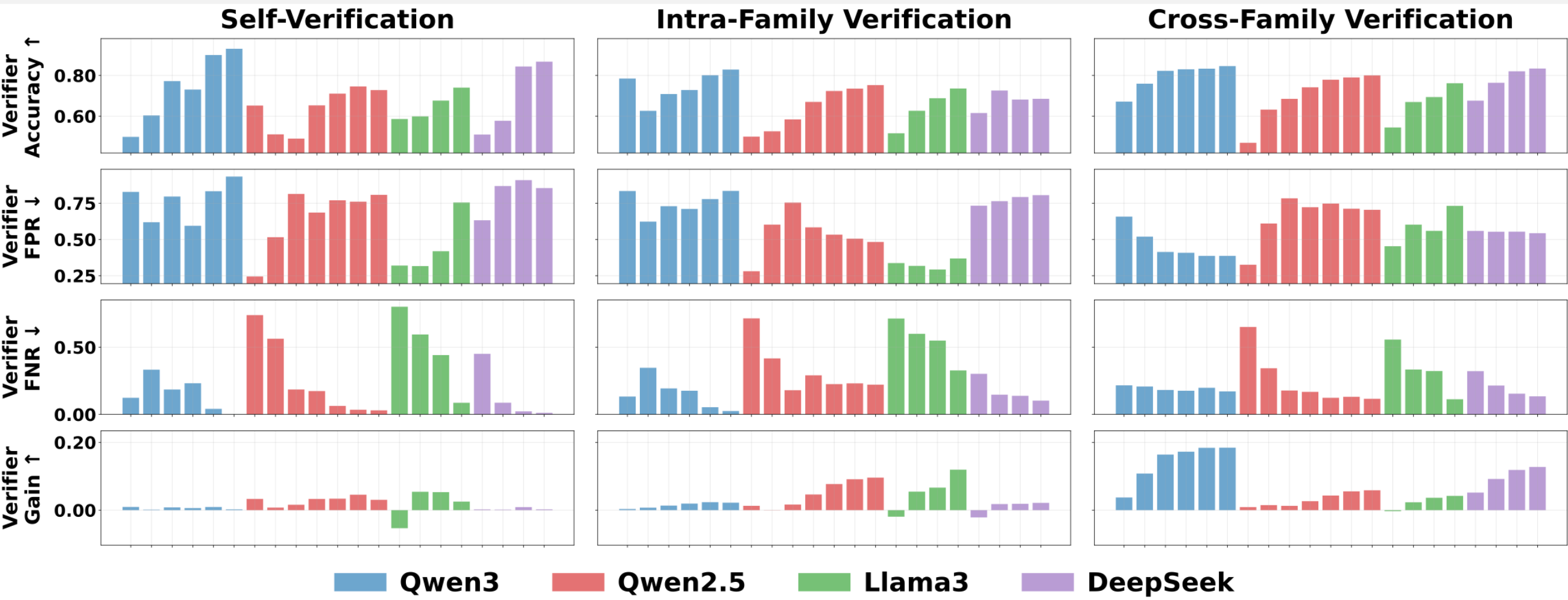
- Verifier gain tracks realized improvement under rejection sampling, with tighter alignment at larger sampling budgets.
- Verifier gain can be estimated from a single verification round without running expensive rejection sampling.

#3 Better Solvers Are Not Better Self-Improvers

- In self- and intra-family settings, stronger verifiers are more likely to accept incorrect solutions.
- Verification accuracy alone does not predict realized test-time improvement.

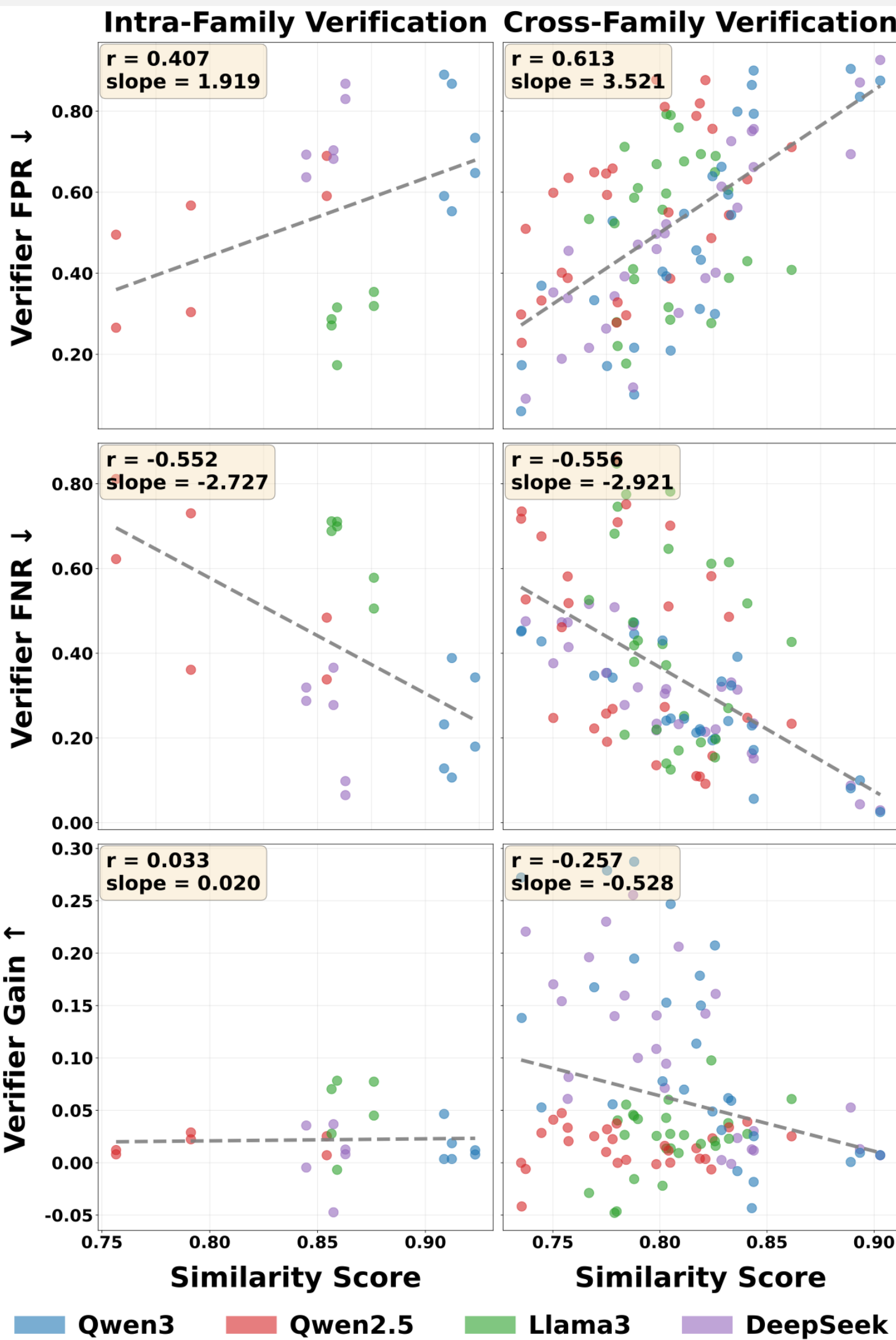
#4 Scaling Model Size Improves Verification Accuracy, Not Self-Improvement

- Larger verifiers yield mixed changes in false positive rate, resulting in little to no self-improvement.
- Strong reasoning models (e.g., Qwen3 and DeepSeek) exhibit little self-improvement due to already performing substantial internal verification.

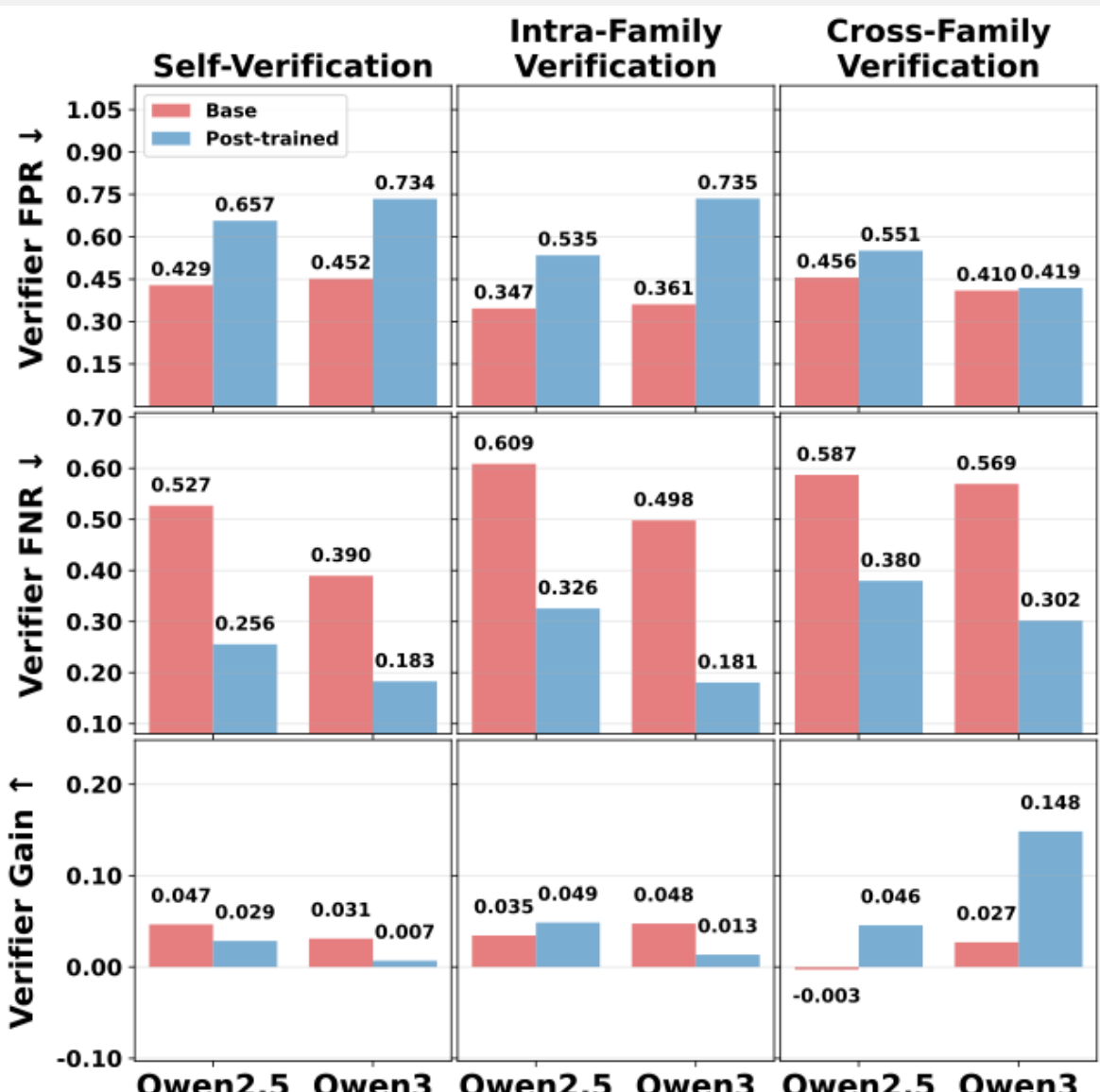
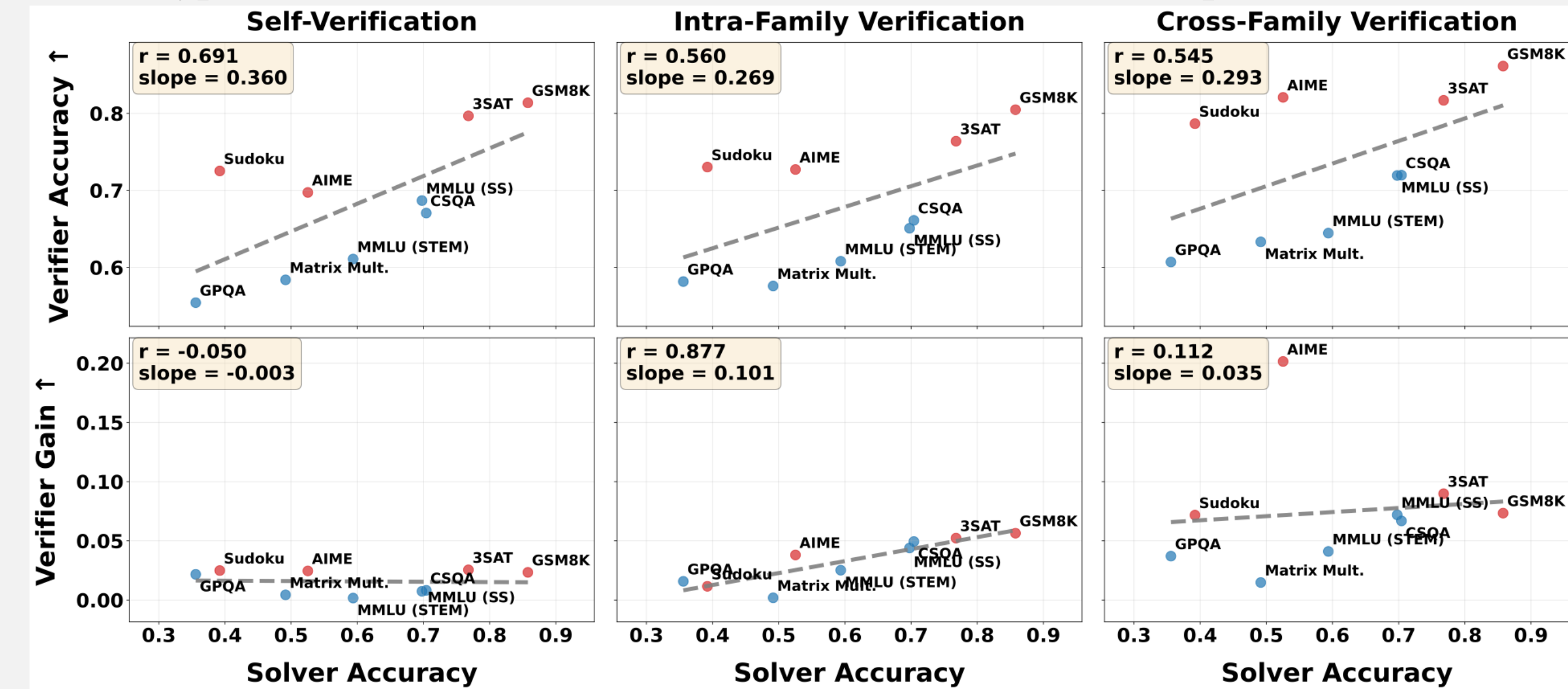


#5 Verifiers Are Biased Toward Solutions That Resemble Their Own

Higher similarity between solver and verifier solution distributions increases the verifier's tendency to accept incorrect solver outputs, reducing test-time improvement.



#6 Task Type and Structure Determine How Much Verification Helps



- Tasks with more verifiable structure receive larger intra-family and cross-family gains, but limited self-improvement.
- Math and logical reasoning benefit most from test-time verification; factual, knowledge-heavy tasks benefit much less.

#7 Reasoning Post-Training Boosts Solving, Yet Weakens Self- and Intra-Family Improvement

Reasoning post-training in Qwen2.5 and Qwen3 increases verification false positives and reduces self-improvement.

Practical Takeaways

- Prefer tasks whose solutions are easier to verify than to derive.
- Use verifier gain, not accuracy, to evaluate a solver-verifier pair.
- Prefer verifiers whose reasoning differ from those of the solver.
- Avoid relying on strong reasoning models as their own verifiers.